

Primary sources & further reading

Part 04 — 📖 Large Language Models · 04-llm/papers/PAPERS.md · 04-llm/resources/RESOURCES.md

Papers & Primary Sources — Large Language Models

arXiv links are canonical and stable. Read the three starred papers in order before anything else; the rest are reference reading organised by the phase of the pipeline they cover.

The canon (read in this order)

#	Paper	Year	Why it matters	Link
1	Attention Is All You Need — Vaswani et al.	2017	The transformer. You studied it in Module 03 [03-nlp]; reread it knowing what it becomes.	arXiv:1706.03762
2	★ Language Models are Few-Shot Learners (GPT-3) — Brown et al.	2020	Scale turns next-token prediction into in-context learning. The paper that made this module necessary.	arXiv:2005.14165
3	★ Training Compute-Optimal LLMs (Chinchilla) — Hoffmann et al.	2022	The 20-tokens-per-parameter rule, and the correction that reshaped every subsequent training run.	arXiv:2203.15556
4	★ Scaling Laws for Neural Language Models — Kaplan et al.	2020	Loss as a power law in N, D, C. The reason labs bet \$100M before a run starts.	arXiv:2001.08361

Post-training & alignment

Paper	Year	One-line takeaway	Link
Training language models to follow instructions (InstructGPT)	2022	SFT + RLHF; the template every chat model still follows	arXiv:2203.02155
Direct Preference Optimization (DPO)	2023	Skip the reward model; a classification loss on preference pairs	arXiv:2305.18290
Constitutional AI — Bai et al.	2022	RLAIF: the model critiques itself against written principles	arXiv:2212.08073
LIMA: Less Is More for Alignment	2023	1,000 curated examples beat 50k noisy ones; quality over quantity in SFT	arXiv:2305.11206
Training a Helpful and Harmless Assistant with RLHF — Anthropic	2022	The practical RLHF write-up: data collection, reward modelling, KL details	arXiv:2204.05862

Efficient inference & adaptation

Paper	Year	One-line takeaway	Link
LoRA: Low-Rank Adaptation — Hu et al.	2021	Freeze W, learn $dW = BA$; the default fine-tuning method	arXiv:2106.09685
QLoRA — Dettmers et al.	2023	4-bit base + LoRA: fine-tune 70B on one GPU	arXiv:2305.14314
Fast Inference from Transformers via Speculative Decoding — Leviathan et al.	2022	Draft-and-verify: 2-3x decode speedup with an unchanged output distribution	arXiv:2211.17192
FlashAttention — Dao et al.	2022	Exact attention, IO-aware tiling; made long context affordable	arXiv:2205.14135
GQA: Grouped-Query Attention — Ainslie et al.	2023	Shrink the KV cache 4-8x with negligible quality loss	arXiv:2305.13245
Efficient Memory Management for LLM Serving with PagedAttention (vLLM)	2023	KV cache as virtual memory; the throughput unlock in modern serving	arXiv:2309.06180

Failure modes & evaluation-adjacent

Paper	Year	One-line takeaway	Link
Lost in the Middle — Liu et al.	2023	Long-context retrieval accuracy dips in the middle; position your evidence deliberately	arXiv:2307.03172
The False Promise of Imitating Proprietary LLMs — Gudibande et al.	2023	Why distilling from a chat model's outputs does not recover its capability	arXiv:2305.15717
Holistic Evaluation of Language Models (HELM)	2022	Multi-metric evaluation; the honest answer to "how good is this model"	arXiv:2211.09110

How to read an LLM paper (30 minutes, 3 passes)

1. **Pass 1 (5 min)**: abstract, Figure 1/Table 1, conclusion. What's the claim, and at what scale?
2. **Pass 2 (15 min)**: experimental setup, ablations, baseline fairness. Was the comparison compute-matched?
3. **Pass 3 (10 min)**: limitations and what would break at a different scale. LLM results are notoriously scale-fragile in both directions.

Keep a one-paragraph note per paper. Prices, leaderboards and SOTA claims expire; mechanisms don't.

Resources — Large Language Models

Courses & lectures (free)

Resource	Why	Link
Karpathy — Neural Networks: Zero to Hero	Build GPT from scratch; the "Let's build GPT" and "Tokenizers" lectures are the core of this module	https://karpathy.ai/zero-to-hero.html
Karpathy — Intro to Large Language Models	1-hour overview of the pretrain -> finetune pipeline	https://karpathy.ai/
Stanford CS336 — Language Modeling from Scratch	The rigorous version: data, systems, scaling, alignment	https://stanford-cs336.github.io/
Hugging Face LLM Course	Transformers, tokenizers, fine-tuning, with runnable notebooks	https://huggingface.co/learn/llm-course
Hugging Face RLHF book (blog chapters)	The clearest free RLHF/DPO walkthrough	https://huggingface.co/blog/rlhf
Jay Alammar — The Illustrated Transformer	The diagrams everyone borrows	https://jalammar.github.io/illustrated-transformer/

Repositories worth cloning

Repo	What's inside
https://github.com/karpathy/nanoGPT	A readable GPT training pipeline in ~300 lines
https://github.com/karpathy/llm.c	The same idea in pure C/CUDA, for seeing the metal
https://github.com/vllm-project/vllm	The serving engine the industry converged on; read the PagedAttention docs
https://github.com/ggerganov/llama.cpp	Quantized local inference; the GGUF/q4 formats in the notes live here
https://github.com/huggingface/trl	Post-training reference implementations: SFT, DPO, RLHF
https://github.com/huggingface/peft	LoRA/QLoRA/adapters as an API
https://github.com/EleutherAI/lm-evaluation-harness	The open eval harness; pairs with Module 15
https://github.com/Lightning-AI/litgpt	Pretraining/fine-tuning recipes with reproducible configs

Where to watch the landscape (not read, watch)

Source	What it gives you
https://huggingface.co/open-llm-leaderboard	Open-model quality comparison; read the methodology caveats
https://artificialanalysis.ai/	Price/latency/quality curves across providers — the Module 11 selection data
https://github.com/rasbt/LLMs-from-scratch	Raschka's book repo; the patient from-scratch build

Cheat sheets

- OpenAI API reference — sampling parameters and their real semantics:
<https://platform.openai.com/docs/api-reference/chat>
- Hugging Face chat templating (`apply_chat_template`):
https://huggingface.co/docs/transformers/main/en/chat_templating
- vLLM docs (quantization, batching, speculative decoding): <https://docs.vllm.ai/>
- Also see `../../_shared/cheatsheets/`

Communities

- `r/LocalLLaMA` — the self-hosting crowd; best place to learn what actually fits on which GPU
- Hugging Face forums — <https://discuss.huggingface.co/>
- EleutherAI Discord — the open-research community behind GPT-NeoX and the eval harness